

CPS Enrollment Forecasting — Feature Dictionary

What every model input means, how it is built, and why it is leak-free · one row per **school × grade × school year**

The goal. Predict next year's enrollment for each school & grade. Two models split the work: a **GBT** specialist for grades 1–8 & 10–12, and an **XGBoost** specialist for the entry grades K & 9.

The golden rule (no leakage). Every feature uses only information available **before** the year being predicted — last year, two years ago, or a survey from a prior year. Nothing from the target year is ever used.

Reading the fill rate. It is the share of rows that have a real value. `*_AVG_3YR` features are ~100% because they use whatever history exists (even just one year). `*_2YR_AGO` and `*_TREND` need a second year of history, so they sit near 84–89%. Anything still missing is filled with the training-set median for the GBT, and kept as-is for XGBoost (which handles blanks natively).

1 · Same-grade history strongest signal

The same school & grade in earlier years — by far the best predictor of next year.

Feature	What it means	How it is built	Fill
SAME_GRADE_LAST_YEAR	How many students were in this exact school & grade last year .	Last year's enrollment for this school+grade.	100%
SAME_GRADE_2YR_AGO	The same count two years ago .	Enrollment for this school+grade two years back.	83.6%
SAME_GRADE_AVG_3YR	The school+grade's typical size recently — smooths out one odd year.	Average of the last up to 3 years (uses fewer if that's all there is).	100%
SAME_GRADE_TREND	Is this grade growing or shrinking ? Positive = grew, negative = shrank.	Last year minus two years ago.	83.6%

2 · Feeder-grade history cohort flow

The grade just below, in earlier years — the cohort that "promotes up" into this grade.

Feature	What it means	How it is built	Fill
FEEDER_GRADE_LAST_YEAR	Size of the grade one below at this school last year — they mostly move up into this grade.	Last year's enrollment of grade N–1 at the same school (falls back to this grade's own last-year count if there is no grade below).	100%
FEEDER_GRADE_2YR_AGO	The same feeder count two years ago .	Grade N–1 at this school, two years back.	84.2%
FEEDER_GRADE_TREND	Is the feeder cohort growing or shrinking ?	Feeder last year minus feeder two years ago.	84.2%

3 · Cohort survival rate promotion strength

What fraction of a feeder grade actually shows up in the next grade the following year. CPS's classic method.

Feature	What it means	How it is built	Fill
COHORT_SURVIVAL_RATE	How well students carried over from the feeder grade into this grade — e.g. 0.95 means 95%	This grade ÷ last year's feeder grade, then shifted back one	84.1%

	moved up.	year so it only uses the past.	
AVG_SURVIVAL_RATE_3YR	The steady-state carry-over rate, averaged so one weird year doesn't dominate.	Average of that ratio over the last 3 years.	100%

4 • Whole-school size school context

The total size of the whole school in earlier years — captures schools that are broadly growing or closing down.

Feature	What it means	How it is built	Fill
SCHOOL_TOTAL_LAST_YEAR	Total students in the whole school last year (all grades).	Sum of last year's enrollment across every grade in the school.	100%
SCHOOL_TOTAL_2YR_AGO	The whole-school total two years ago .	School-wide total, two years back.	85.0%
SCHOOL_TOTAL_AVG_3YR	The school's typical overall size recently.	Average of the school total over the last up to 3 years.	100%
SCHOOL_TOTAL_TREND	Is the whole school growing or shrinking?	School total last year minus two years ago.	85.0%

5 • District-wide grade trend system context

How this grade is doing across **all of CPS** — picks up city-wide demographic shifts (e.g. fewer kindergartners citywide).

Feature	What it means	How it is built	Fill
DISTRICT_GRADE_ENROLLMENT_LAST_YEAR	Total students in this grade across every CPS school last year.	Sum of last year's enrollment for this grade district-wide.	100%
DISTRICT_GRADE_ENROLLMENT_2YR_AGO	The district-wide grade total two years ago .	District grade total, two years back.	88.9%
DISTRICT_GRADE_ENROLLMENT_AVG_3YR	The typical district-wide size of this grade recently.	Average of the district grade total over the last up to 3 years.	100%
DISTRICT_GRADE_ENROLLMENT_TREND	Is this grade growing or shrinking across the whole district ?	District grade total last year minus two years ago.	88.9%

6 • 5Essentials school-climate scores school quality

Five survey-based ratings of school quality (each 1–5), from the prior year. A good school holds onto families. Missing years are filled with that school's own earlier average before any district-wide fallback.

Feature	What it means	How it is built	Fill
EFFECTIVE_LEADERS_LAST_YEAR	Prior-year rating of school leadership (1–5).	Last year's 5Essentials pillar score (1–5) for the school, attached with a one-year lag. Blanks filled with the school's own earlier-year average.	92.8%
COLLABORATIVE_TEACHERS_LAST_YEAR	Prior-year rating of teachers working together (1–5).		92.8%
INVOLVED_FAMILIES_LAST_YEAR	Prior-year rating of family engagement (1–5).		92.8%

SUPPORTIVE_ENVIRONMENT_LAST_YEAR	Prior-year rating of a safe, supportive environment (1–5).		92.8%
AMBITIOUS_INSTRUCTION_LAST_YEAR	Prior-year rating of challenging, high-quality teaching (1–5).		92.8%

7 · Supporting inputs (not in the fill-rate list)

Used by the models or the pipeline, but not part of the "model features" fill-rate table above.

Field	What it means	Role	
ENROLLMENT	The actual number of students — the thing we predict .	Target (label), not an input.	—
GRADE	The grade level (K, 1, ..., 12).	Lets the GBT tell grades apart (encoded internally).	—
IS_MIGRANT_ANOMALY_YEAR	Flags 2022–2024, the migrant-arrival surge.	Those years are down-weighted in training (counted less); also an XGBoost input.	—

Generated for the CPS Enrollment Forecasting pipeline · feature table `ml_enrollment_feature_table_v4` . Fill rates are from the most recent Stage 1 build. All features are lagged or prior-year, so the models never see the year they are predicting.